INCH-POUND

MIL-C-70585B (AR)
9 August 1991
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, DATA TRANSMISSION (PIVADS): 9397510

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements and tests for the Fire Control Processor Data Transmission circuit card assembly, referred to as the Data Transmission Board (DTB). The DTB is a 7 1/4 by 11 inch, double layered, multiwire circuit card. The Data Transmission Board contains three digital data links that provide interface capability between the Fire Control Processor (FCP) and external devices for testing and training purposes. The data link circuitry is not utilized in normal fire control operations (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Test and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360906	-	Interface Device Adapter Board D, Printed Wiring Assembly
9397497	-	Cable Assembly W1
9397510	-	Data Transmission Assembly
9397512	-	Data Transmission Board Schematic
TPS12554986	-	Test Program Set Manual

SPECIAL PACKAGING INSTRUCTIONS

PS9397510 - Packaging of Data Transmission Assembly

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE/INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018)

ELECTRONICS INDUSTRIES ASSOCIATION

RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Application for copies should be addressed to the Electronics Industries Association, 345 East 47th St. New York NY 10017)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9397510, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the DTB shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.2.2 ANSI/IPC-DW-425. IPC-DW-425/11 Type C1 plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration. The DTB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 Hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The DTB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The DTB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The DTB shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The DTB shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg. F and +155 deg. F.

3.3.4.2 Non-operating (storage). The DTB shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg. F for a period of 24 hours and +160 deg. F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The DTB shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram 9397512.

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3.5 Performance.

3.5.1 Functional description. The DTB shall contain three digital data links that provide interface capability between the FCP and external devices for testing, training, and meterological parameter insertion purposes. The data link circuitry shall not be utilized in normal day-to-day fire control operations. Data link #1 shall include the Development Test/Operational Test (DTOT) data link and the Training data link. The DTOT and Training data links shall not operate simultaneously. The First-In First-Out (FIFO) memory and interface hardware are shared by the DTOT and Training links and are software controlled. The DTOT link is unidirectional and the Training link is bidirectional. Data link #2 is bidirectional and shall be identified as the Test data link for development testing support. Bidirectional data link #3 shall be identified as the Spare (future weather station) data link. The Test and Spare data links shall not be operational at the same time and shall utilize a single Universal Asynchronous Receiver Transmitter (UART). This is accomplished by multiplexing the inputs and outputs of the two links to the single UART. The multiplexing is under software control. A 308 kHz differential clock signal shall be output by a line driver form the DTB for the Spare data link. The DTB shall also contain non-volatile random access memory (NVRAM) in two 2 k byte EEPROM chips for storage and retrieval of ballistic parameters by the FCP. One chip shall contain the lower 8 bits (lower byte) of data to form a 16-bit data word. The DTB shall also include connections and control hardware for adding two additional 32 k byte EPROMs. A set of six relays are provided to activate circuits external to the weapon system when certain system switches and indicators are activated. One of these relays is for the MILES interface. A second set of contacts on these relays shall put relay status on the data bus for functionally testing the relays and for outputting system switch and indicator status onto the data links.

3.5.2 Power circuit.

3.5.2.1 DTB power requirements without EPROMs U13 and U14 installed. The DTB shall require +5 Vdc for the integrated circuits and +24 Vdc for the relays per Table I.

TABLE I. INPUT POWER REQUIREMENT

Input Voltage (in Vdc)	Input Connections (Polarity)	Connector-Pins	Current (in amperes)	Ripple (in mV p-p)
+4.75 to +5.25	(+5 V)	P1-32,33,34,72,73,74	3 Max.	100 Max.
	(+5 V RTN)	P1-35,36,37,75,76,77		
+22 to +30	(+24 V(BAT))	P2-34	1.2 Max.	100 Max.
	(+24 V(BAT)RTN)	P2-69		

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3.5.2.2 DTB power requirements with EPROMs U13 and U14 installed. The DTB shall require power per 3.5.2.1 and, when populated with the U13 and U14 EPROM chips, the DTB shall also require +5.0 ± .25 Vdc applied to inputs +5 V EPROM VCC at P2-35 and +5 V EPROM VPP at P2-70 with both returns connected to +5 V RTN per Table I. +5 V EPROM VPP may be connected to +5 V EPROM VCC. The maximum current drawn by EPROMs U13 and U14 together shall be:

	From +5 V EPROM VCC	From +5 V EPROM VPP
Read/Standby:	250 mA	10 mA

3.5.2.3 Voltage detector circuit. When ±4.80 ± 0.05 = Vdc is applied to the +5 V input of the DTB at connector P1 per Table I with READ (P1-24) and RESET (P1-29) logic high, $\overline{OE}$ (J1-34) shall remain low until the +5 V input is +4.5 to +4.75 Vdc. $\overline{OE}$ shall then go and remain high.

3.5.3 Clock circuit. With power applied to the DTB per 3.5.2.1 and an 8 MHz ± 1,176 Hz TTL square wave clock signal applied to LS CLOCK (8 MHz) at P1-70, the output at test point CLOCK (J1-50) shall be an 8 MHz ± 1,176 Hz TTL square wave inverted with respect to the input at P1-70. The TTL outputs at test points QD, DEC, CNTR RESET and U CLK shall be per the timing diagram of Figure 1. QD shall have a frequency of 615.385 kHz ± 91 Hz. U CLK shall have a frequency of 307.692 kHz ± 46 Hz. U CLK shall be applied to differential driver U55 and be available at DATA LINK #3 CLOCK (308 kHz) with P2-10 (HI) and P2-45 (LO).

3.5.4 Control circuit. With power applied to the DTB per 3.5.2, applying each of the DTB input groups of the left-hand column of Table II shall activate the corresponding on-board control line shown in the right-hand column of Table II.

TABLE II. Control circuit logic

Input Connector-Pin	Logic Level	Output Connector-Pin	Logic
P1-64	L	J1-39	L
P2-31	L	J2-1	L
P1-31	L	J1-49	H
P1-24	L	J2-2	L
P1-24	L	J1-46	H
P1-52	L	J2-50	L
P1-52	L	J1-45	H

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P1-66	L	J1-44	H
P1-28	L	J1-47	H
P1-28	L	J1-48	H
P1-29	L	J1-29	H
P1-29	L	J1-34	L
P1-29	L	J1-18	L
P1-29	L	J2-6	H
P1-29	L	J2-16	H
P1-29	L	J2-26	H
P1-29	L	J2-36	H
P1-29	L	J2-4	H
P1-29	L	J2-31	H
P1-29	L	J2-32	H
P1-29	L	J2-43	H
P1-29	L	J2-42	H
P1-66	L	J1-17	L
P2-31	L	J1-17	L
P2-31 and P1-52	L L	P1-12	L
P1-66 and P1-52	L L	P1-12	L
INTERNAL GROUND		P1-30	L

3.5.5 Expansion EPROMs read operation. Sockets for U13 and U14 are available for future EPROM expansion. No requirement currently exists for components.

3.5.6 Read/write decoders.

3.5.6.1 Read decoder operation. With power applied to the DTB per 3.5.2 and with the logic levels of Table III applied to I/O BOARD 2 SELECT, READ, 2 DELAY WRITE, B1/2 A16,14,13, and 12, the

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outputs of read decoder U17 shall be enabled low as shown in Table III. Data bus buffers U3 and U4 shall also set to the "read" direction (from D0 through D15, to B1/2 D0 through B1/2 D15).

TABLE III. Read decoder logic

DTB Inputs							Read Decoder
$\overline{\text{I/O BOARD 2 SELECT}}$ (P1-66)	$\overline{\text{READ}}$ (P1-24)	$\overline{\text{2 DELAY WRITE}}$ (P1-64)	B1/2 A16 (P1-13)	B1/2 A14 (P1-14)	B1/2 A13 (P1-54)	B1/2 A12 (P1-15)	Output Enabled
L	L	H	L	L	L	L	$\overline{\text{EEPROM READ}}$ (J1-30)
L	L	H	L	L	L	H	N/A
L	L	H	L	L	H	L	N/A
L	L	H	L	L	H	H	$\overline{\text{F STATUS READ}}$ (J1-27)
L	L	H	L	H	L	L	$\overline{\text{FIFO READ}}$ (J1-27)
L	L	H	L	H	L	H	$\overline{\text{STATUS READ}}$ (J1-26)
L	L	H	L	H	H	L	$\overline{\text{UART READ}}$ (J1-25)
L	L	H	L	H	H	H	$\overline{\text{RELAY STATUS READ}}$ (J1-23)

NOTES: H = LOGIC HIGH
L = LOGIC LOW

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3.5.6.2 Write decoder operation. With power applied to the DTB per 3.5.2 and with the logic levels of Table IV applied to I/O BOARD 2 SELECT, 2 DELAY WRITE, READ, B1/2 A16, 14, 13, and 12, the outputs of write decoder U16 shall be enable low as shown in Table IV. Data bus buffers U3 and U4 shall also be set to the "write" direction (from B1/2 DO through B1/2 D15, to DO through D15).

TABLE IV. Write decoder logic.

DTB Inputs							Read Decoder
$\overline{\text{I/O BOARD 2 SELECT}}$ (P1-66)	$\overline{\text{2 DELAY WRITE}}$ (P1-64)	$\overline{\text{READ}}$ (P1-24)	B1/2 A16 (P1-13)	B1/2 A14 (P1-14)	B1/2 A13 (P1-54)	B1/2 A12 (P1-15)	Output Enabled
L	L	H	L	L	L	L	$\overline{\text{EEPROM WRITE}}$ (J1-24)
L	L	H	L	L	L	H	N/A
L	L	H	L	L	H	L	N/A
L	L	H	L	L	H	H	$\overline{\text{RESET 1}}$ (J1-22)
L	L	H	L	H	L	L	$\overline{\text{FIFO WRITE}}$ (J1-21)
L	L	H	L	H	L	H	N/A
L	L	H	L	H	H	L	$\overline{\text{UART LOAD}}$ (J1-20)
L	L	H	L	H	H	H	$\overline{\text{UART LATCH WRITE}}$ (J1-19)

NOTES: H = LOGIC HIGH
L = LOGIC LOW

3.5.7 End-Around-Test (EAT) of the address bus. With power applied to the DTB per 3.5.2 and with I/O BOARD 2 SELECT (P1-66) and READ (P1-24) low, ADDRESS BUS READ (J1-31) shall be decoded low. Alternately applying 8000 then FFFF to address bus lines A16 through A1 of B1/2 shall cause 8000 then FFFF to appear on the data bus lines D15 through DO of B1/2.

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3.5.8 Non-Volatile Random Access Memory (NVRAM) write/read. With power and clock applied to the DTB, data on data bus B1/2 shall be capable of being written to NVRAM or read from NVRAM and put back onto data bus B1/2. The NVRAM shall hold its programmed data with DTB power and clock off. The programmed data shall be readable when power and clock are again applied to the DTB. Address lines A1-A11 shall select the memory location for each data word (2,048 possible) to be written to read from NVRAM.

3.5.8.1 NVRAM (EEPROM) write operation. To write a data word to NVRAM, the following conditions and logic levels must be met:

a. Power is applied to the DTB per 3.5.2.
b. Address lines B1/2 A1 through B1/2 A11 (P1-61, 21, 60, 20, 59, 19, 57, 17, 56, 16, and 55, respectively) are set to the desired memory word location to write into.
c. Data lines B1/2 DO through B1/2 D15 (P1-48, 8, 47, 7, 46, 6, 45, 5, 44, 4, 43, 3, 42, 2, 41, and 1, respectively) are set to the desired data word for writing into NVRAM.
d. RESET (P1-29) is high.
e. I/O BOARD 2 SELECT (P1-66) is low.
f. 2 DELAY WRITE (P1-64) is low.
g. B1/2 A16 (P1-13) is low.
h. B1/2 A14 (P1-14) is low.
i. B1/2 A13 (P1-54) is low.
j. B1/2 A12 (P1-15) is low.
k. READ (P1-24) is high.
l. UDS (P1-68) is low.
m. LDS (P1-28) is low.

The following test points shall have the logic levels indicated:

a. RESET (J1-29) is low.
b. CE HI (J1-32) is low.
c. CE LO (J1-33) is low.
d. UDS (J1-48) is high.
e. LDS (J1-47) is high.
f. CE (J2-3) is high.
g. EEPROM WRITE (J1-24) is low.
h. EEPROM READ (J1-30) is high.
i. READ (J2-2) is high.
j. READ (J1-46) is low.
k. BOARD SELECT (J1-44) is high.
l. 2 DELAY WRITE (J1-39) is low.
m. A16 (J2-12) is low.
n. A14 (J2-13) is low.
o. A13 (J2-12) is low.
p. A12 (J2-11) is low.
q. BUS ENABLE (J1-17) is low.

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3.5.8.2 NVRAM (EEPROM) read operation. The following conditions and logic levels shall be met to read a data word from NVRAM and put it on data bus lines B1/2 D0 through B1/2 D15 at P1-48, 8, 47, 7, 46, 6, 45, 5, 44, 4, 43, 3, 42, 2, 41, and 1, respectively:

a. Power and clock are applied to the DTB per 3.5.2 and 3.5.3, respectively.
b. Address lines B1/2 A1 through B1/2 A11 (P1-61, 21, 60, 20, 59, 19, 57, 17, 56, 16, and 55, respectively) are set to the desired memory word location to be read.
c. B1/2 A16 (P1-13) is low.
d. B1/2 A14 (P1-14) is low.
e. B1/2 A13 (P1-54) is low.
f. B1/2 A12 (P1-15) is low.
g. $\overline{\text{UDS}}$ (P1-68) is low.
h. $\overline{\text{LDS}}$ (P1-28) is low.
i. $\overline{\text{2 DELAY WRITE}}$ (P1-64) is high.
j. $\overline{\text{READ}}$ (P1-24) is low.
k. I/O BOARD 2 SELECT (P1-66) is low.
l. Two rising clock edges have occurred since $\overline{\text{I/O BOARD 2 SELECT}}$ (P1-66) was pulled low to allow for chip enable access time.

The following test points shall have the logic levels indicated:

a. A16 (J2-14) is low.
b. A14 (J2-13) is low.
c. A13 (J2-12) is low.
d. A12 (J2-11) is low.
e. UDS (J1-48) is high.
f. $\overline{\text{LDS}}$ (J1-47) is high.
g. $\overline{\text{2 DELAY WRITE}}$ (J1-39) is high.
h. $\overline{\text{READ}}$ (J2-2) is low.
i. READ (J1-46) is high.
j. $\overline{\text{BOARD SELECT}}$ (J1-44) is high.
k. $\overline{\text{EEPROM READ}}$ (J1-30) is low.
l. $\overline{\text{CE}}$ (J2-3) is high.
m. $\overline{\text{CE HI}}$ (J1-32) is low.
n. $\overline{\text{CE LO}}$ (J1-33) is low.
o. $\overline{\text{EEPROM WRITE}}$ (J1-24) is high.
p. $\overline{\text{BUS ENABLE}}$ (J1-17) is low.
q. $\overline{\text{OE}}$ (J1-34) is low.

3.5.9 Operation of link #1 as the unidirectional DTOT data dump.

a. $\overline{\text{RUN DTOT}}$ (P2-11) shall be grounded to +5 V RTN. The low on $\overline{\text{RUN DTOT}}$ will call up FCP software related to the DTOT dump function when the UART status byte is read if certain other weapon system conditions are also met simultaneously.

b. Power and clock shall be applied to the DTB per 3.5.2 and 3.5.3, respectively.

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c. Shift register U44, the FIFOs and all DTB UARTs shall reset when power is applied to the DTB or RESET 1 is decoded low per 3.5.6.2 for at least 400 nanoseconds with power applied.

d. With $\overline{\text{F STATUS READ}}$ decoded low per 3.5.6.1, the UART status bits and link run signals shall be read out on the data bus per Table V. A data word can travel through the FIFOs within 4 microseconds, but it requires approximately 286 microseconds (11 bits 38.4 kHz) for the UARTs to transmit a single word. For this reason, weapon system software checks status bits TBRE and TRE for both the upper and lower bytes before sending a data byte to the FIFOs for UART transmission. This is done to ensure the UARTs are capable of loading another byte for transmission. The number of bytes that can be loaded are as follows:

D11(TBRE, upper byte) or D3 (TBRE, lower byte)	D10(TRE, upper byte) or D2 (TRE, lower byte)	Meaning
L	L	UART has 2 bytes...no data can be loaded
L	H	Transient state; within one microsecond, will go to the HL state
H	L	UART has 1 byte of data one byte can be loaded
H	H	UART has no data...up to 2 bytes can be loaded

e. With $\overline{\text{FIFO WRITE}}$ decoded low per 3.5.6.2, pulsing $\overline{\text{DELAY 2}}$ (P1-52) low shall cause system software to load (byte by byte) 16-bit data words representing critical system parameters onto data bus lines B1/2 DO through B1/2 D15 and follow each byte with a positive Shift-In (SI, J1-36) pulse to input each byte into the FIFO buffer. The Shift-In pulses in conjunction with FIFO architecture shall load, shift, propagate and stack the data bytes into the 128 word FIFO buffer. (The 128 word buffer consists of two 64 word stages. The DTB reset shall have caused the Shift-Outlines (SO1, 2, 3, 4, at J2-6, 16, 25, 35) of the first stage to be high. When data has propagated to the outputs of the first stage, the Output Ready signals (OR-1,2,3,4 at J2-5, 15, 25, 35) from the

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first stage shall act as Shift-In pulses for the second stage and the data shall continue to propagate to the outputs of the second stage). The pulsing of DELAY 2 shall also cause Data Acknowledge (DTACK) pulses to be generated and output at P1-12.

f. The first data word loaded into the FIFO shall appear on FIFO output lines TD0 through TD15 (J2-40, 39, 38, 37, 30, 29, 28, 27, 20, 19, 18, 17, 10, 9, 8, and 7, respectively) within 4 microseconds after the second byte's Shift-In pulse returns low. Output Ready (OR, J1-37) shall go high.

g. With the upper and lower byte UART Transmitter Registers empty, TRE1 at J2-33 and TRE2 at J2-34 shall be high. With TRE1, TRE2 and OR high, NEXT (J1-38) shall go high.

h. The high NEXT signal, applied to shift register U44, shall cause the Transmitter Buffer Register Load signal (TBRL, J1-40) to pulse low, followed by a positive Shift-Out (SO, J1-41) pulse. TBRL, when low, shall load the data word on TD0 through TD15 into the Transmitter Buffer Registers of UARTs U34 and U35. The low to high transition of the trailing edge of the TBRL pulse shall transfer the data bits from the UART's Transmitter Buffer Registers to the UART's Transmitter Registers where they shall be clocked out serially. The upper data byte shall be clocked out TRO1 (J2-31) and CPU TO DATA LINK #1 HI DATA at P2-1, 2 (HI) and P2-36, 37 (LO) complete with one start bit, 8 data bits, an odd parity bit and one stop bit per Figure 2. The lower data byte shall be clocked out TRO2 (J2-32) and CPU TO DATA LINK #1 LO DATA at P2-3, 4 (HI) and P2-38, 39 (LO) complete with one start bit, 8 data bits, an odd parity bit and one stop bit per Figure 2. The UART clock must be 16 times the data rate, therefore, with a 615 kHz clock, the upper and lower bytes shall each be transmitted out simultaneously at approximately 38.4 kHz (615 kHz 16). The Shift-Out (SO) pulse shall cause the second data word loaded into the FIFO to be output on FIFO output lines TD0 through TD15. The Output Ready (OR) line shall again pulse high and, when the first data word has been completely sent out of the UARTs, TRE1 and TRE2 shall again go high. The NEXT signal will again go high and trigger shift register U44 to generate a UART load pulse (TBRL), followed by a Shift-Out (SO) pulse.

i. The cycle shall repeat until all of the data words loaded into the FIFO buffer have been serially transmitted from the DTB. FCP software quickly parallel loads the FIFO buffer. The data stored in the FIFO buffer may then be clocked out serially to an external device at a slower rate without further intervention by the FCP software.

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TABLE V. Data bit flags in the status word of link #1

BIT	MEANING
D15	FE, Framing Error (upper byte)
D14	OE, Overrun Error (upper byte)
D13	PE, Parity Error (upper byte)
D12	DR, Data Received (upper byte)
D11	TBRE, Transmitter Buffer Register Empty (upper byte)
D10	TRE1, Transmitter Register Empty (upper byte)
D9	RUN DTOT (P2-11)
D8	RUN TRAINING (P2-12)
D7	FE, Framing Error (lower byte)
D6	OE, Overrun Error (lower byte)
D5	PE, Parity Error (lower byte)
D4	DR, Data Received (lower byte)
D3	TBRE, Transmitter Buffer Register Empty (lower byte)
D2	TRE2, Transmitter Register Empty (lower byte)
D1	RUN FCP UNIT TEST (P2-13)
D0	RUN SPARE (P2-14)

3.5.10 Operation of link #1 as the bidirectional training data link. The unidirectional DTOT and bidirectional training data links cannot be operational at the same time because they share common hardware.

a. RUN TRAINING (P2-12) shall be grounded to +5 V RTN. (The low on RUN TRAINING shall cause bit D8 (P1-44) of the data bus to go low when F STATUS READ is decoded low per 3.5.6.1. In a weapon system, the FCP monitors bit D8. A low detected on D8 during F STATUS READ will call the software related to the training data link. Refer to Table V).

b. Power and clock shall be applied to the DTB per 3.5.2 and 3.5.3, respectively.

c. The bidirectional training data link shall reset when power is applied to the DTB or RESET 1 is decoded low per 3.5.6.2 for at least 400 nanoseconds with power applied.

d. Serial data shall be transmitted to the DTB by an external training device over two RS-422 differential lines; one differential line for the upper byte of a 16-bit data word and the other line for the lower byte per Figure 2. Upper byte data shall enter the DTB at P2-58 (HI), P2-59 (LO) and lower byte data shall enter the DTB at P2-60 (HI), P2-61 (LO).

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e. The serial RS-422 data shall be converted to TTL level by line receivers. High logic levels shall be pulled up to +5 volts to satisfy the CMOS UART requirement for a logic high to be greater than or equal to +3 volts.

f. With bit B1/2 D12 (P1-42) high and UART LATCH WRITE of 3.5.6.2 not pulsed, EAT ON (J2-48) shall be high and the 2-line to 1-line data selectors shall pass the received serial data to the UARTs' serial data inputs.

g. The UARTs shall convert the serial data to parallel and the Data Received (DR) flags shall go high (set).

h. FCP software shall poll the UARTs' status bits by decoding F STATUS READ low per 3.5.6.1, detect that RUN TRAINING bit D8 is low and call up the training link software. The FCP shall also detect the Data Received flags are set, check for Parity, Framing and Overrun Errors (PE, FE and OE) and then decode FIFO READ low per 3.5.6.1 even if the data contains errors. FIFO READ, when low, shall reset the Data Received flags and take the UARTs' Receiver Buffer Register output lines out of tristate. The received data shall then be available in parallel form on data bus lines B1/2 D0 through B1/2 D15.

i. FCP software shall read the message (from the training device) presently on the data bus and send a programmed response to the FIFO buffer for transmission back to the external training device. The transmission process and hardware shall be the same as that used by the DTOT dump; paragraphs 3.5.9.d through 3.5.9.i.

3.5.11 Link #1 End-Around-Test (EAT) operation.

a. Power and clock shall be applied per 3.5.2 and 3.5.3, respectively, to the DTB.

b. Link #1 shall reset when power is applied to the DTB or when RESET 1 is decoded low per 3.5.6.2 for at least 400 nanoseconds with power applied. If the end-around-test fails, the DTB must be reset again before continuing.

c. A test data word shall be sent to the FIFO buffer by:

(1) loading the test word onto the data bus, and

(2) pulsing DELAY 2 (P1-52) low with FIFO WRITE (J1-21) decoded low per 3.5.6.2 to generate the Shift-In pulse.

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d. A TBRL (J1-40) pulse shall be generated automatically by clocked shift register U44. When low, TBRL shall load the data word from the FIFO output lines into the UARTs' Transmitter Buffer Registers. The low-to-high transition of TBRL shall transfer the data word from the UARTs' Transmitter Buffer Registers to the UARTs' Transmitter Registers, where the data word shall be serially clocked out TR01 (J2-31) and TR02 (J2-32).

e. With the D12 bit (B1/2 D12 at P1-42) pulled low and with UART LATCH WRITE (J1-19) pulsed (decoded low per 3.5.6.2 then made high), the EAT ON (J2-48) signal shall go low and the 2-line to 1-line data selectors shall route the serially transmitted data word to the Receiver Register Inputs, RRI1 (J1-42) and RRI2 (J1-43), of the UARTs.

f. The Data Received (DR) flags shall set when the serial word has been completely received and converted to parallel. FCP software shall decode F STATUS READ low per 3.5.6.1, when polling to determine UART status. The software shall detect that the DR flags are set and make FIFO READ low per 3.5.6.1 to take the UARTs' Receiver Buffer Register lines out of tristate and reset the DR flags. The data word shall then be available on Data bus lines B1/2 D0 through B1/2 D15.

g. FCP software shall compare the received data word to the original test word which was loaded into the FIFO buffer. The two words shall be identical.

3.5.12 Operation of link #2, the bidirectional test data link. The link #2 requires U13 and U14 EPROMs to be installed. No requirement currently exists for link #2 without these components.

3.5.13 Operation of link #3, the bidirectional spare (weather station) data link. The link #3 requires U13 and U14 EPROMs to be installed. No requirement currently exists for link #3 without these components.

3.5.14 Links #2,3 End-Around-Test (EAT). Link #2 and #3 require U13 and U14 EPROMs to be installed. No requirement currently exists for links #2 and #3 without these components.

3.5.15 UART U51 End-Around-Test (EAT). The end-around-test of UART U51 shall check the transmitting and receiving sections of UART U51, multiplexer U50 and latch U49. The end-around-test shall operate as follows:

a. Power and clock shall be applied to the DTB per 3.5.2 and 3.5.3, respectively.

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b. With power applied to the DTB or RESET 1 decoded low per 3.5.6.2 for at least 400 nanoseconds with power applied, Master Reset (MR, J2-4) shall go high and reset UART U51. TBRE (J2-43) and TRO (J2-42) shall be high within 70 microseconds after a reset pulse is initiated.

c. With STATUS READ decoded low per 3.5.6.1, status flag data bits B1/2 D2(P1-47) and B1/2 D3(P1-7) shall both be high indicating the Transmitter Buffer and Transmitter Registers of UART U51 are both empty.

d. With data bus lines B1/2 D15, D14, and D13 set up high for UART EAT and with all other data bus lines logic low, inputting a UART LATCH WRITE pulse per 3.5.6.2 shall set up latch U49 and latch bits D15, D14, and D13 onto the address lines of multiplexer U50 for routing the transmitted serial data output of the UART U51 around to the receiver input of UART U51.

e. The UART shall convert the received serial data to parallel.

f. Decoding UART READ low per 3.5.6.1 shall load the converted data onto data bus lines B1/2 D0 through B1/2 D7. This received byte shall be identical to the original test byte sent out.

3.5.16 FCP Unit Test.

a. RUN FCP UNIT TEST (P2-13) shall be grounded to +5 V RTN. The low on RUN FCP UNIT TEST will be used to call off -board software related to FCP testing, using the FCP Test Program Set (TPS).

b. Power and clock shall be applied to the DTB per 3.5.2 and 3.5.3, respectively.

c. The DTB shall reset when power is applied or RESET 1 is decoded low per 3.5.6.2 for at least 400 nanoseconds with power applied.

d. With F STATUS READ decoded low per 3.5.6.1, status bit D1(P1-8), which represents RUN FCP UNIT TEST per Table V, shall be detected to be low and the FCP Unit Test software will be called if other external conditions are also met simultaneously.

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3.5.17 Relay circuit operation. With relay returns connected per Table VI, each of the six on-board relays shall activate when the signals of Table VIII are applied. With RELAY STATUS READ decoded low per 3.5.6.1, each activated relay shall pull its associated data bit (per Table VI) low. The normally open, normally closed and common relay contacts, available for off-board use, shall be per Table X. The maximum relay contact current is 1.0 Ampere at 28 Vdc resistive and 0.1 Ampere for a 28 Vdc lamp circuit. Relay coil resistance is 1,560 ohms ±10%.

TABLE VI. Relay operation

Signal Applied (Input)	Bit	Relay Activated	Relay Return
READY TO FIRE LAMP (+24V RTN, P2-33)	D7	K1	+24V, P2-34
TRACK/JAM LAMP (+24V RTN, P2-68)	D6	K2	+24V, P2-34
ACTION SWITCH (+24V, P2-32)	D5	K3	+24V RTN, P2-69
RADIATE COMMAND (+24V, P2-67)	D4	K4	+24V RTN, P2-69
ACQUISITION/TRACK SWITCH (+24V, P2-30)	D3	K5	+24V RTN, P2-69
FIRE TRIGGER RELAY (+24V, P2-65)	D2	K6	+24V RTN, P2-69

3.5.17.1 MILES interface. The MILES interface is the TRACK/JAM relay output at P2-51. The TRACK/JAM relay is energized when the TRACK/JAM/RADIATE lamp in the weapon system is lit and is not energized when the TRACK/JAM/RADIATE lamp is off.

a. With +24 ± Vdc applied to P2-34 with return at P2-68 and P2-69, TRACK/JAM relay K2 shall energize and supply +24 ± Vdc to P2-51 with respect to P2-52.

b. With +24 ± Vdc applied to P2-34 with return at P2-69 only, TRACK/JAM relay K2 shall not energize and the voltage at P2-51 shall be 0 ± 0.1 Vdc with respect to P2-52.

3.5.18 Input/output requirements. DTB inputs shall be as listed in Table VII. DTB outputs shall be as listed in Table VIII.

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3.5.19 TTL voltage levels. The TTL voltage levels for inputs and outputs shall be as follows:

	Min	Typ	Max	Units
VOL	---	0.2	0.4	V
VOH	2.4	3.4	---	V
VIL	---	---	0.7	V
VIH	2.0	---	---	V

3.5.20 RS-422 voltage levels.

3.5.20.1 Differential. The RS-422 clock, line receiver and line driver buffer differential voltage levels shall be as follows:

	High Logic Level	Low Logic Level
Maximum input:	+5.5 V	-5.5 V
Sensitivity:	$V_{iH}=V_{in}-\overline{V_{in}}>+0.05$ V	$V_{iL}=V_{in}-\overline{V_{in}}<-0.5$ V

3.5.20.2 Common mode. The RS-422 common mode voltage levels shall be:

	Min	Typ	Max	Units
VOL	---	---	0.45	V
VOH	2.5	3.2	---	V
VIL	---	---	0.8	V
VIH	2.0	---	---	V

3.5.21 Spare connector pins. The DTB shall contain spare pins on connectors J1, J2, P1, and P2 as listed in Tables IX and X.

3.5.22 Test points. The DTB shall contain test points as listed in Tables XI and XII.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

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TABLE VII. Inputs

CONNECTOR /PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-1	B1/2 D15	TTL
P1-2	B1/2 D13	TTL
P1-3	B1/2 D11	TTL
P1-4	B1/2 D9	TTL
P1-5	B1/2 D7	TTL
P1-6	B1/2 D5	TTL
P1-7	B1/2 D3	TTL
P1-8	B1/2 D1	TTL
P1-9	+5 v RTN	GROUND
P1-13	B1/2 A16	TTL
P1-14	B1/2 A14	TTL
P1-15	B1/2 A12	TTL
P1-16	B1/2 A10	TTL
P1-17	B1/2 A8	TTL
P1-18	+5 v RTN	GROUND
P1-19	B1/2 A6	TTL
P1-20	B1/2 A4	TTL
P1-21	B1/2 A2	TTL
P1-24	$\overline{\text{READ}}$	TTL
P1-28	$\overline{\text{LDS}}$	TTL
P1-29	$\overline{\text{RESET}}$	TTL
P1-32	+5 V	+5 V
P1-33	+5 V	+5 V
P1-34	+5 V	+5 V
P1-35	+5 V RTN	GROUND
P1-36	+5 V RTN	GROUND
P1-37	+5 V RTN	GROUND
P1-41	B1/2 D14	TTL
P1-42	B1/2 D12	TTL
P1-43	B1/2 D10	TTL
P1-44	B1/2 D8	TTL
P1-45	B1/2 D6	TTL
P1-46	B1/2 D4	TTL
P1-47	B1/2 D2	TTL
P1-48	B1/2 D0	TTL
P1-49	$\overline{\text{+5 v RTN}}$	GROUND
P1-52	$\overline{\text{DELAY 2}}$	TTL
P1-53	B1/2 A15	TTL
P1-54	B1/2 A13	TTL
P1-55	B1/2 A11	TTL
P1-56	B1/2 A9	TTL
P1-57	B1/2 A7	TTL
P1-58	+5 V RTN	GROUND
P1-59	B1/2 A5	TTL
P1-60	B1/2 A3	TTL
P1-61	B1/2 A1	TTL

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TABLE VII. Inputs. (Continued)

CONNECTOR /PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-64	2 DELAY WRITE	TTL
P1-66	I/O BOARD 2 SELECT	TTL
P1-68	UDS	TTL
P1-70	LS CLOCK (8MHz)	
P1-72	+5 V	+5 V
P1-73	+5 V	+5 V
P1-74	+5 V	+5 V
P1-75	+5 V RTN	GROUND
P1-76	+5 V RTN	GROUND
P1-77	+5 V RTN	GROUND
P2-6	+5 V RTN	GROUND
P2-7	DATA LINK #2 TO CPU DATA HI	RS-422
P2-8	DATA LINK #3 TO CPU DATA HI	RS-422
P2-11	RUN DTOT	TTL
P2-12	RUN TRAINING	TTL
P2-13	RUN FCP UNIT TEST	TTL
P2-14	RUN SPARE	TTL
P2-15	RUN DATA LINK #2	TTL
P2-29	A18	TTL
P2-30	ACQUISITION/ TRACK SW ENABLE	CV OR 24 V
P2-31	1 MEG SELECT	TTL
P2-32	ACTION SWITCH	OV OR 24
P2-33	READ TO FIRE LAMP	OV OR 24 V
P2-34	+24 V (BAT)	POWER
P2-35	+5 V EPROM VCC	POWER
P2-41	+5 V RTN	GROUND
P2-42	DATA LINK #2 TO CPU DATA LO	RS-422
P2-43	DATA LINK #3 TO CPU DATA LO	RS-422
P2-52	+24 V (BAT) RTN	POWER
P2-58	DATA LINK #1 TO CPU HI DATA HI	RS-422
P2-59	DATA LINK #1 TO CPU HI DATA LO	RS-422
P2-60	DATA LINK #1 TO CPU LO DATA HI	RS-422
P2-61	DATA LINK #1 TO CPU LO DATA LO	RS-422
P2-64	A17	TTL
P2-65	FIRE TRIGGER RELAY	OV OR 24 V
P2-66	A19	TTL
P2-67	RADIATE COMMAND	OV OR 24 V
P2-68	TRACK/JAM LAMP	OV OR 24 V
P2-69	+24 V (BAT) RTN	POWER
P2-70	+5 V EPROM VPP	POWER

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TABLE VIII. Outputs

CONNECTOR /PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-12	DTACK	TTL
P1-30	I/O BOARD 2 IN	TTL
P1-65	INTERRUPT 3	TTL
P2-1	CPU TO DATA LINK #1 HI DATA HI	RS-422
P2-2	CPU TO DATA LINK #1 HI DATA HI	RS-422
P2-3	CPU TO DATA LINK #1 LO DATA HI	RS-422
P2-4	CPU TO DATA LINK #1 LO DATA HI	RS-422
P2-5	CPU TO DATA LINK #2 DATA HI	RS-422
P2-9	CPU TO DATA LINK #3 DATA HI	RS-422
P2-10	DATA LINK #3 CLOCK (308 KHz) HI	RS-422
P2-16	READY TO FIRE COM	(RELAY)
P2-17	READY TO FIRE NO	(RELAY)
P2-18	READY TO FIRE NC	(RELAY)
P2-19	ACTION SWITCH COM	(RELAY)
P2-20	ACTION SWITCH NO	(RELAY)
P2-21	ACTION SWITCH NC	(RELAY)
P2-22	ACQUISITION/TRACK SWITCH COM	(RELAY)
P2-23	ACQUISITION/TRACK SWITCH NO	(RELAY)
P2-24	ACQUISITION/TRACK SWITCH NC	(RELAY)
P2-25	FIRE TRIGGER RELAY COM	(RELAY)
P2-26	FIRE TRIGGER RELAY NO	(RELAY)
P2-27	FIRE TRIGGER RELAY NC	(RELAY)
P2-36	CPU TO DATA LINK #1 HI DATA LO	RS-422
P2-37	CPU TO DATA LINK #1 HI DATA LO	RS-422

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TABLE VIII. Outputs (Continued)

CONNECTOR /PIN	SIGNAL NAME	DESCRIPTIVE DATA
P2-38	CPU TO DATA LINK #1 LO DATA LO	RS-422
P2-39	CPU TO DATA LINK #1 LO DATA LO	RS-422
P2-40	CPU TO DATA LINK #2 DATA LO	RS-422
P2-44	CPU TO DATA LINK #3 DATA LO	RS-422
P2-45	DATA LINK #3 CLOCK (308 KHz) LO	RS-422
P2-51	TRACK/JAM	(RELAY)
P2-54	RADIATE COMMAND COM	(RELAY)
P2-55	RADIATE COMMAND NO	(RELAY)
P2-56	RADIATE COMMAND NC	(RELAY)

TABLE IX. Connector P1 Spare Pins

CONNECTOR PIN
10
11
22
23
25
26
27
31
38
39
40
50
51
62
63
67
69
71

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TABLE X. Connector P2 Spare Pins

PIN/DESIGNATION
28
46
47
48
49
50
53
57
62
63

TABLE XI. Connector J1 Test Points

PIN	SIGNAL NAME
1	D0
2	D1
3	D2
4	D3
5	D4
6	D5
7	D6
8	D7
9	D8
10	D9
11	D10
12	D11
13	D12
14	D13
15	D14
16	D15
17	BUS ENABLE
18	RESET
19	UART LATCH WRITE
20	UART LOAD
21	FIFO WRITE
22	RESET 1
23	RELAY STATUS READ
24	EEPROM WRITE
25	UART READ

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TABLE XI. Connector J1 Test Points (Continued)

PIN	SIGNAL NAME
26	STATUS READ
27	FIFO READ
28	F STATUS READ
29	RESET
30	EEPROM READ
31	ADDRESS BUS READ
32	CE HI
33	CE LO
34	OE
35	EPROM OE
36	SI
37	OR
38	NEXT
39	2 DELAY WRITE
40	TBRL
41	SO
42	RRI 1
43	RRI 2
44	BOARD SELECT
45	DELAY 2
46	READ
47	LDS
48	UDS
49	1 MEG SELECT
50	CLOCK

TABLE XII. Connector J2 Test Points

PIN	SIGNAL NAME
1	1 MEG SELECT
2	READ
3	CE
4	MR
5	OR1
6	SO1
7	TD15
8	TD14
9	TD13
10	TD12
11	A12
12	A13
13	A14
14	A16
15	OR2

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TABLE XII. Connector J2 Test Points (continued)

PIN	SIGNAL NAME
16	SO2
17	TD11
18	TD10
19	TD9
20	TD8
21	CNTR RESET
22	DEC
23	QD
24	UCLK
25	OR3
26	SO3
27	TD7
28	TD6
29	TD5
30	TD4
31	TR01
32	TR02
33	TRE1
34	TRE2
35	OR4
36	SO4
37	TD3
38	TD2
39	TD1
40	TD0
41	RRI3
42	TR0
43	TBRE
44	UART LOAD
45	MUX CONTROL C
46	MUX CONTROL B
47	MUX CONTROL A
48	EAT ON
49	INT 3 CONTROL
50	DELAY 2

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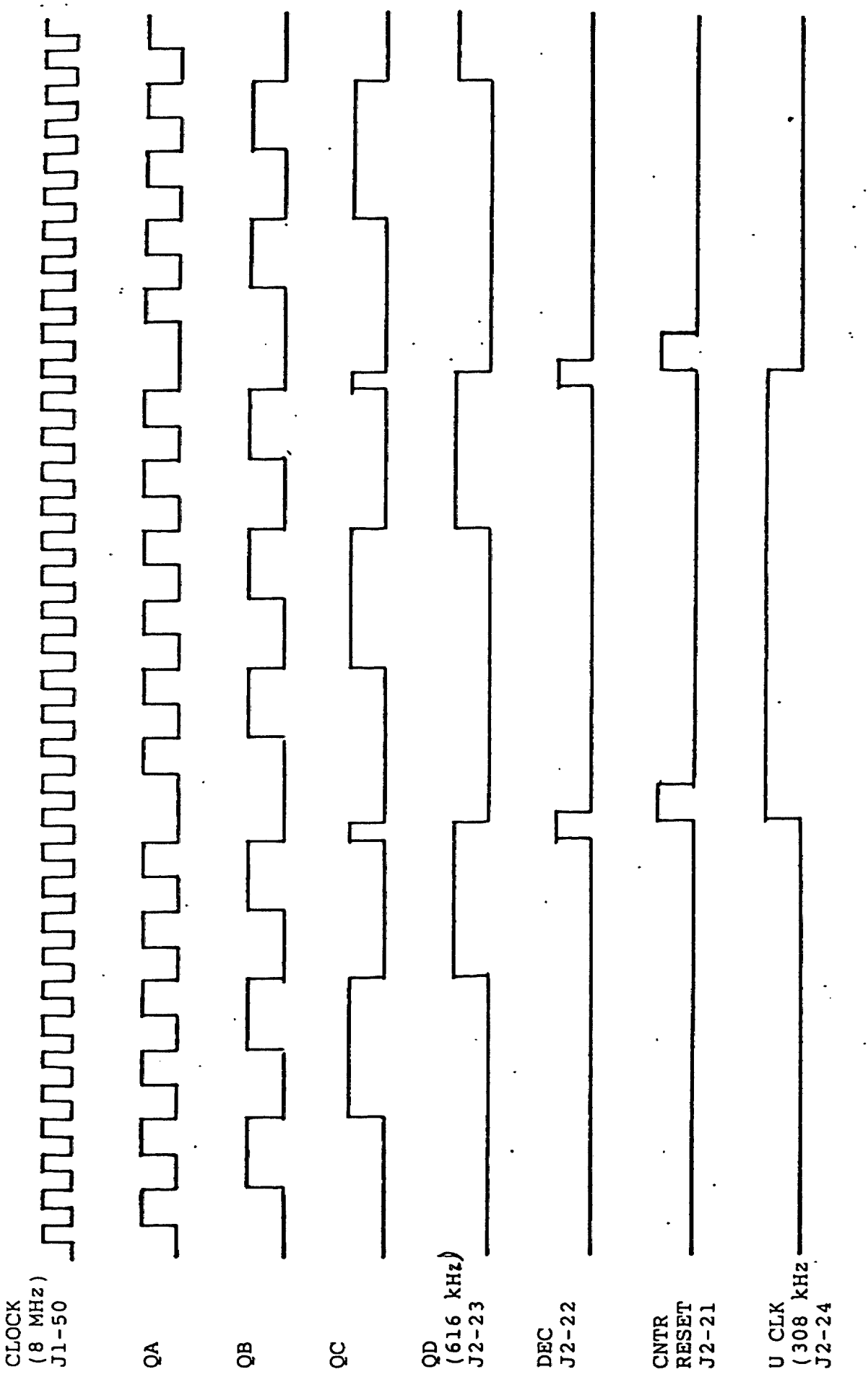
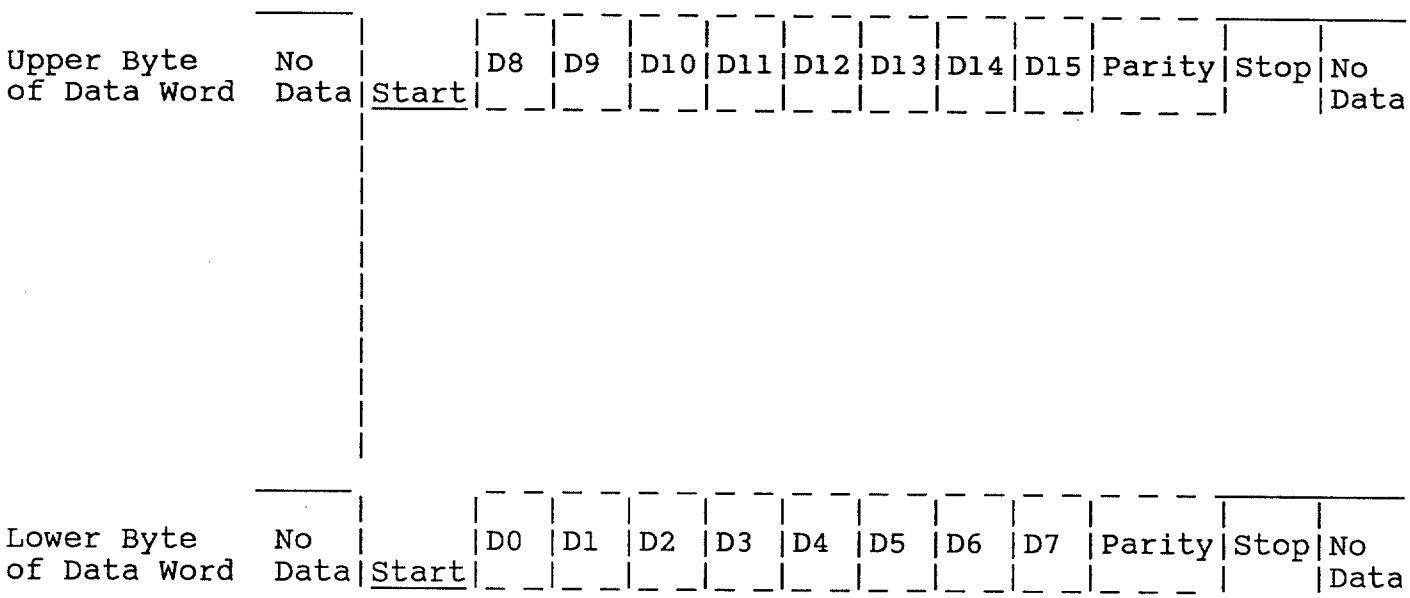


FIGURE 1. Clock circuit timing diagram.

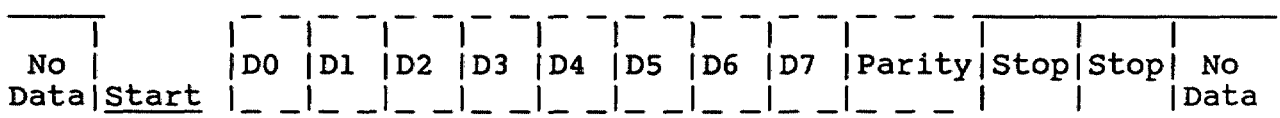
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NOTES:
1. The data lines are always high when no data bytes are being received or transmitted.
2. Each start bit is always low and each stop bit is always high.
3. Received bytes are checked for odd parity. All transmitted parity bits are odd.
4. The upper and lower data bytes are serially transmitted from the DTB simultaneously.

FIGURE 2. Data link #1; input and output data format

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NOTES:
1. The data lines are always high when no data bytes are being received or transmitted.
2. Each start bit is always low and each stop bit is always high.
3. Received bytes are checked for odd parity. All transmitted parity bits are odd.

FIGURE 3. Data links #2,3; input and output data format

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Special sampling inspection (see 4.5) (if required).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) DTBs for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XIII (Functional Test) and Table XIV (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XIII shall be performed on a 100% basis.

TABLE XIII. Functional Tests
(100% Inspection)

Characteristic	Requirement	Test Procedure
Critical:	None	
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	

4.5 Special sampling. One DTB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XIII and shall then meet the tests in Table XIV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIV. Special sampling

Characteristic	Requirements	Test Procedure
Critical:	None	
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Amplitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table XV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7. The following items are required to test the DTB board and shall be assembled at the AN/USM-465A test station prior to test execution. Proper operation of interface devices and test equipment shall be verified before using same to run diagnostic tests.

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TABLE XV. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Interface Device	9360906 (GFE)
2.	Cable IDC W1	9397497 (GFE)
3.	Digital Card Tester	AN/USM-465A
4.	ComCode Program Tape	Version 2.0
5.	Test Program Set Manual	TPS 12554986 (GFE)
6.	Test Program Set Tape	TPT 12554986 (GFE)
7.	Oscilloscope	Input impedance of 1 Mohm or greater, single sweep capability, with probe, ±100 Vdc and 500 Hz capability

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The DTB shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the DTB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

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d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the DTB shall tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.3 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 502 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The DTB shall be tested in accordance with the requirements of section 3.5.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS9397510.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The data transmission circuit card assembly shall be used in the fire control processor in the PIVADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now cognizance of ARDEC.

6.5 Subject term (key word) listing.

Processor, Fire Control
Product Improved Vulcan Air Defense System (PIVADS)
Test Program Set (TPS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A657)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-70585B (AR)	2. DOCUMENT DATE (YYMMDD) 9 August 1991

3. DOCUMENT TITLE
CIRCUIT BOARD, DATA TRANSMISSION (PIVADS): 9397510

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander US Army ARDEC	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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